

No evidence that compliance assistance reduces environmental violations

Lessons from “The Effect of Compliance Assistance on Pollution Discharges and Violations of Environmental Regulations” by Paul J. Ferraro & Jay P. Shimshack in *Journal of Policy Analysis and Management*, 2026.

In new research studying hundreds of polluting facilities that were offered free compliance assistance workshops and training, **offering polluting facilities free help to follow the rules didn’t make them more likely to follow them.**

BACKGROUND

Environmental enforcement in the U.S. has long centered on inspecting facilities, catching violations, and levying fines. **Compliance assistance** takes a different approach: regulators offer workshops, technical guidance, and other support to help facilities understand and meet regulatory requirements. This approach reflects the belief that many violations result from confusion or limited expertise rather than deliberate corner-cutting. Compliance assistance programs have become increasingly popular as a policy lever, but evidence on whether they are effective is scarce.

In a new study, Ferraro and Shimshack (2026) find that compliance assistance workshops and training did not reduce regulatory violations or pollution discharges. The researchers

analyzed data from hundreds of Ohio wastewater treatment facilities that had recently been cited for violations and were invited to attend workshops rolled out over several months.

THE CHALLENGE

Evaluating whether compliance assistance programs are effective is difficult to do in practice. Simply comparing the behavior of facilities before vs. after receiving the training risks attributing program success to changes in behavior that are actually driven by other factors.

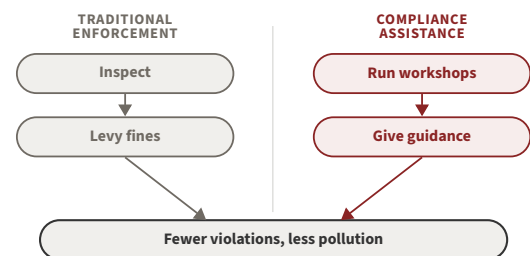
Comparing facilities that signed up against those that never did also raises some concerns. The facilities that sign up for a workshop might be different from those that don’t in ways that have nothing to do with the workshop, like a manager who cares about reducing violations. Comparing attendees and non-attendees might simply reflect who chose to walk through the door, not what they learned once inside.

To avoid these concerns, **Ferraro and Shimshack compared facilities that received compliance assistance and training early to facilities that had signed up but not yet received the training.** Arguably, these facilities would have had similar motivations and similar histories. The absence of differences between these facilities prior to receiving the training provides support for this.

NO EFFECT ON VIOLATIONS OR DISCHARGES

Compared with not-yet-trained facilities, trained facilities violated environmental regulations just as often and discharged just as much pollution. That held within facilities as well as across them: tracking the same facilities over time, their violations did not decline in the months after they attended a workshop (see figure). Ferraro and Shimshack find no support for the expectation that compliance assistance and training reduce violations or pollution discharges. **At least in this context, a lack of understanding at these facilities does not appear to be what was holding compliance back.**

TWO APPROACHES TO COMPLIANCE



This study tests whether compliance assistance works.

THE WORKSHOPS' EFFECT ON VIOLATIONS, MONTH BY MONTH

Source: Ferraro & Shimshack (2026), Fig. 2a.



Note: This figure shows the estimated effect of compliance assistance workshops on the number of facility violations. Each dot shows whether facilities had more (above the zero line) or fewer (below the zero line) violations in a given month relative to the workshop month (vertical dashed line). The estimates account for each facility's usual number of violations, the local weather, and any common trends shared across facilities; the shaded band shows the range of estimates within the 95% confidence interval. If the workshops had an effect on violations, the line would drop below zero after the workshop.

IMPLICATIONS FOR POLICY

If these results hold in other contexts – beyond Ohio wastewater treatment plants – they strongly caution against leaning on compliance assistance to reduce violations and pollution. The evidence that inspections and penalties work is stronger. With enforcement budgets already shrinking, **funding a compliance assistance program out of that same budget could mean more violations and more pollution.**

One program that didn't work isn't a verdict on compliance assistance everywhere. Rather, **this study highlights the value of rigorous program evaluation.** It also points to an opportunity for researchers and regulators to work together: by piloting new programs and measuring their effects before rolling them out widely, researchers can help policymakers learn what works, advancing evidence-based policy making.

FOR MORE INFORMATION

Contact Jay Shimshack (jay.shimshack@virginia.edu) with questions about this research.

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